

University of Economics in Prague

Faculty of Informatics and Statistics



# **User Interface Concept for Smart Glasses**

## **BACHELOR THESIS**

Program: Applied Informatics

Field of Study: Multimedia in the Economic Practice

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Prague, May 2020



## **Declaration**

I declare that I have written this bachelor thesis “User Interface Concept for Smart Glasses” on my own, using reliable sources I have referred to below.

Prague, 6 May 2020

.....

Robert Poláček

## **Acknowledgement**

I would like to thank my supervisor Martin Kopta for his readiness and helpful guidance, my bachelor thesis leader Zdeněk Vondra for his support and patience, and my parents Martin and Vladana for their comprehension of my extended studies.

## **Abstract**

The goal of this thesis is to design a concept of the user interface for smart glasses for everyday personal use, that should help developers to make digital products for this platform.

The theoretical part contains an analysis of manufacturers' approaches to user interfaces inside smart glasses. The first section is a list of models, currently available in the market. The second section aims to collect important information, that helps to design the user interface concept (elements' placement in user's point of view, user experience evolution and user needs). The third section aims to summarize current visual design approaches in extended reality industry.

The practical part is the concept of the user interface itself, based on the research from the theoretical part. The concept includes optimal ways to control smart glasses, their physical characteristics, user interface components, suggested apps list and visual design recommendations.

## **Keywords**

smart glasses, design, concept, user interface, UI, graphic user interface, GUI, zero UI, no UI, user experience, UX, digital design, augmented reality, AR, mixed reality, MR, augmented virtuality, AV, virtual reality, VR, extended reality, XR, hybrid reality, merged reality, operating system, OS, launcher, hologram, eyewear device

## **JEL Classification**

n/a

## **Anotace**

Cílem této práce je vypracovat návrh uživatelského rozhraní chytrých brýlí pro každodenní osobní využití, který by měl vývojářům pomoci tvořit digitální produkty pro tuto platformu.

Teoretická část obsahuje rozbor výrobců chytrých brýlí a jejich přístupů k uživatelskému rozhraní. První sekce je seznam modelů, které jsou v současné době dostupné na trhu. Druhá sekce je zaměřena na shromáždění důležitých informací, které pomohou vytvořit návrh uživatelského rozhraní (umístění prvků v okolí uživatele, evoluce uživatelské zkušenosti a potřeby koncového uživatele). Třetí sekce je zaměřena na shrnutí současných výtvarných přístupů v oblasti rozšířené reality.

Praktická část obsahuje samotný návrh uživatelského rozhraní na základě výzkumu z teoretické části. Návrh zahrnuje vhodné způsoby ovládání chytrých brýlí, jejich fyzické vlastnosti, prvky uživatelského rozhraní, seznam základních aplikací a doporučení ohledně výtvarného pojetí.

## **Klíčová slova**

chytré brýle, tvorba, návrh, uživatelské rozhraní, grafické uživatelské rozhraní, uživatelská zkušenost, digitální design, rozšířená realita, smíšená realita, rozšířená virtualita, virtuální realita, VR, hybridní realita, sloučená realita, operační systém, OS

## **JEL klasifikace**

n/a

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# Introduction

Currently, we use augmented reality in our smart phones. But was this technology really meant to be used on them? I believe that smart phones are only transitional platform to smart glasses. What is more, Apple believes, that glasses might replace smartphones one day (Hughes, 2019). This prediction might seem impossible from today's perspective, but there is a current race among technological industry leaders, who strive to hit the market with personal smart glasses, that will become bestseller and spark the smart glasses industry (Haselton, 2019).

In this bachelor thesis, I want to understand this incoming trend and be part of it in its early stages. With my knowledge of digital design I would like to come up with user interface for smart glasses.

Due to the current quarantine situation, I am not allowed to test my ideas using real smart glasses. The results I came up with are based on online sources, primary research and experimenting using the smart glasses prototype.

The reason why I write this thesis in English is that I want my concept to be available to the open source community, for designers all over the world, who struggle with creating user interface or other digital products for smart glasses. This concept can also be used or finished by any developer, who looks for visual representation of his or her operating system or subsequent launcher.

This bachelor thesis is released with Creative Commons License 3.0 (type CC-BY-SA). It means that it can be shared, used, customized or remixed for free. If it would be shared in any resemblance, it must be with my name and under the same license type.

In addition, it may help me to get a job abroad, if there would be any foreign studio or individual enthusiastic about my work.

# **1 Theoretical Part**

## 1.1 Basic Terms and Abbreviations

While I have been cutting through smart glasses market I found out that sources, even the very credible ones such as dictionaries (Lenovo) or even manufacturers such as Lenovo (TechTarget, 2018) or Apple (Apple) do not use proper term designations – often replacing MR and AV with AR term, which means totally different kind of visual experience.

In my opinion, it might be an intention to customize current term to absorb all forms of hybrid realities, because *“Terms are the fundamental building blocks of communication that drives the global economy.”* (Warburton) But it might result to chaos in proper designations.

For educational purpose, I adapted foreign (Flavian, et al., 2018) graphic system for proper understanding of basic “reality” terms. Also I illustrated the realities for better understanding of the differences among them. The illustrations were created with free stock images by (Andy) and (Deckert) from The Noun Project.

### 1.1.1 Realities Definition

**Reality (R)** = Our physical reality that we live in every day. No device needed.

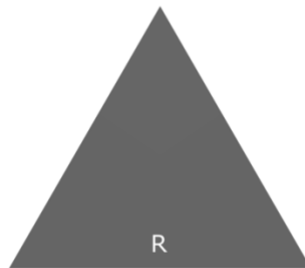


Figure 1 (Poláček, 2020)



Figure 2 (Poláček, 2020)

**Virtual Reality (VR)** = Completely digital world without limits. To enter, you have to wear special device for replacing sensory perception (Irvine, 2017).

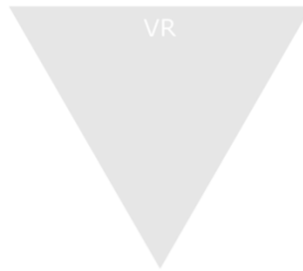


Figure 3 (Poláček, 2020)

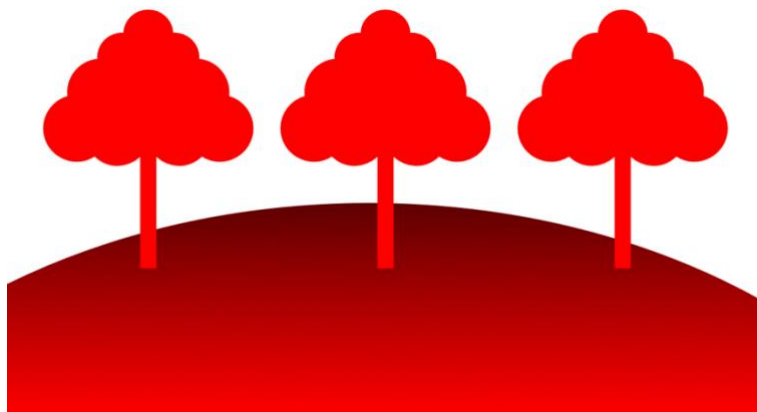


Figure 4 (Poláček, 2020)

Combination of these two realities creates another reality.

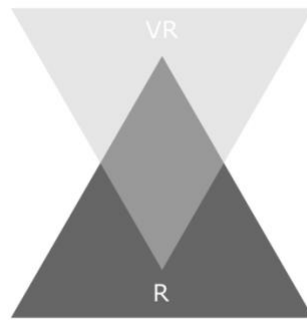


Figure 5 (Poláček, 2020)

**Augmented Reality (AR)** = It is the lightest connection of reality with virtual world. The reality is enhanced by simple virtual elements. The elements cannot be placed to your physical surrounding and cannot be interacted with them. (Irvine, 2017)

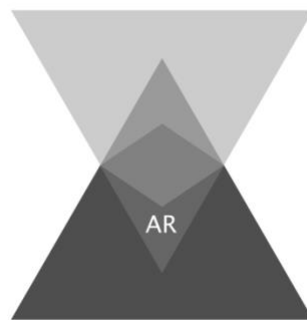


Figure 6 (Poláček, 2020)



Figure 7 (Poláček, 2020)

**Mixed Reality (MR)** = It is the middle intersection between physical world and virtual reality. The reality is enhanced by complex virtual elements. Your surrounding blends with it and you can interact with the virtual objects. (Irvine, 2017)

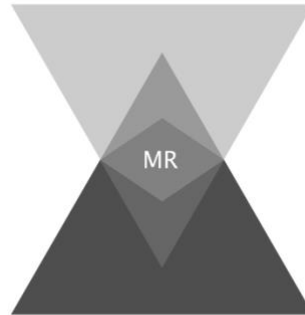


Figure 8 (Poláček, 2020)



Figure 9 (Poláček, 2020)

**Augmented Virtuality (AV)** = It is the last type of environment connected with the real world. The virtual reality is enhanced by real (physical) elements. (Flavian, et al., 2018)

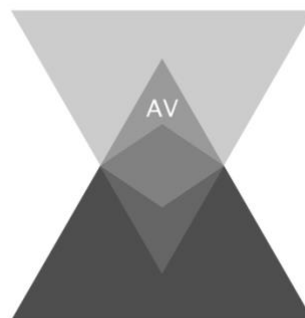


Figure 10 (Poláček, 2020)



Figure 11 (Poláček, 2020)

**Extended Reality (XR)** = Term that includes every version of digitally enhanced reality, namely AR, MR, AV and VR. (Irvine, 2017)

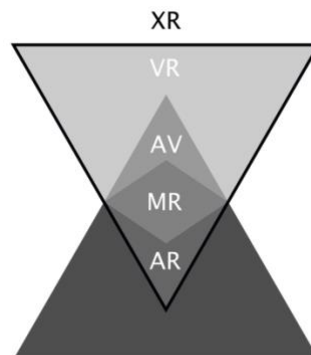


Figure 12 (Poláček, 2020)

**Merged Reality** = This term was first introduced by Intel to describe very intense virtual reality for project Alloy (Intel, 2016). Nowadays people usually confuse it with Mixed Reality (Tait, 2016).

**Hybrid Reality** = this term is rarely used and its definition differs. It is usually used for either AR, MR or AV (TechTarget, 2018) but there is no exact standardized definition.

**Hyper Reality** = a reality beyond VR, where simulation is undetectable (e.g. Matrix). (Soules)

## 1.1.2 Other Terms

**User Experience (UX)** = *“The overall experience of a person using a product, especially in terms of how easy or pleasing it is to use.”* (Oxford Dictionary)

**User Interface (UI)** = *“The means by which the user and a computer system interact, in particular the use of input devices and software.”* (Oxford Dictionary)

**Graphic User Interface (GUI)** = A user interface where user uses graphic elements to control the device.

**Zero User Interface (Zero UI, No UI)** = A screen-less user interface (user can control the device using e.g. natural means of human communication, for example voice or gestures. (Lundquist, 2015)

**Ambient Interface** = A type of Zero UI. The device controls itself by physical variables' values from surrounding.

**Operating System (OS)** = „A software that communicates with the hardware and allows other programs to run.” (TecTerms)

**Launcher** = “A software that causes applications to load and run. Launching an app is naturally built into every operating system; however, third-party app launchers are available and popular for mobile devices. An app launcher replaces the stock user interface for organizing the home screen and app icons.” (The Computer Language Company) Launcher is dependent on OS capabilities and cannot change only some of the OS' components.

**Haptic Feedback** = A perceptible physical response after executed action (e.g. vibrations after pressing a button).

**Degrees of Freedom (DoF)** = Possible movements of holograms' manipulation.

**Field of Vision (FoV)** = An area in front of your eyes, where you can see virtual content.

**Hologram** = A 3D virtual object placed in real environment.

**Eyewear Device** = Any technological device that can be worn over eyes.

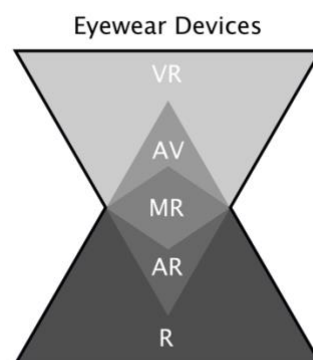


Figure 13 (Poláček, 2020)

**Smart Glasses** = “A pair of glasses that contain computer technology so that, for example, they can be used in a similar way to a smartphone.” (Cambridge University Press)

- Despite the “pair” definition of smart glasses, Temple-Mounted Display eyewear devices (e.g. Glasses by Google) are considered as smart glasses too. In my opinion, it is because the “smart glasses” term is catchier and more comprehensible for general public.
- MR has currently form of headsets, but soon they might have form of glasses.

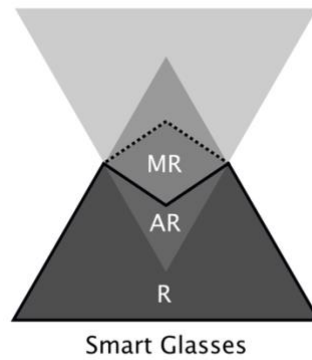


Figure 14 (Poláček, 2020)

## 1.2 Smart Glasses Models

To start with, there is enormous amount of smart glasses models in the market. I tried to focus on the ones, that have been often mentioned in various credible websites' ranks. But while doing the research, I figured out that the ranks are often misleading (probably using a weak methodology). They include very early development stage models or models, that had been canceled years before the rank was published (Cherdo, 2020) (Jaehnig, 2020).

So, I have to go through all smart glasses models I found in the selected time period of approximately two weeks. Although there can be only a limited description of a model, it is because the rest is not important for the thesis.

There can be also no ranking system delivering the best smart glasses model. The glasses have a lot of important parameters, that no general parameter can be the one preferred. In this list, the sorting criterium is based on a "reality blending score". In other words, the list is sorted by how much of virtual presence is in the glasses. At the end of this section, there is a section called

Summary of Smart Glasses Models that highlights the models to prefer for the following design process.

### ***Spectacles by Snap***

First of all, it is important to say that Spectacles do not augment reality in any virtual experience. It does not allow users to perceive the world differently. Neither through audio, nor through visuals. It is still a smart glasses model though, because it has complex digital features, that cannot be found in any everyday glasses.

The current model Spectacles 3 is basically plain personal sunglasses that can take 3D photos by pressing a button. After taking a photo, you can pair them with Snapchat app to export the photos to your phone. Additionally, there is a notification diode that notifies a new message in Snapchat app in your smartphone. (Snap)

## 1.2.1 Augmented Reality Glasses

Augmented reality might augment (enhance) reality by different factors. Although the thesis focuses on glasses providing audiovisual experience, some big companies start with audio experience-based models. It is possible, that visual extension might be added to these glasses in future.

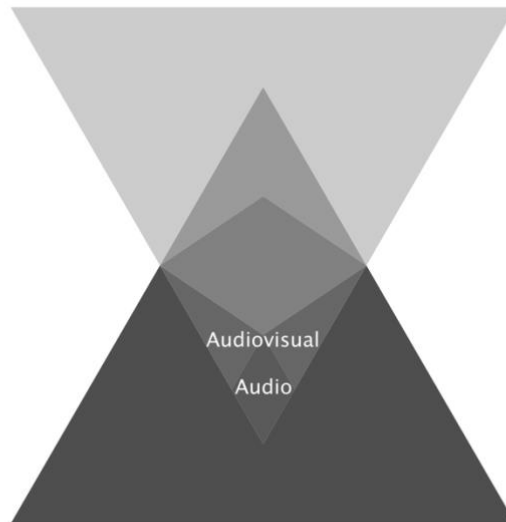


Figure 15 (Poláček, 2020)

### 1.2.1.1 Audio Models

The first experience of virtual world, delivered by smart glasses, can be made through sound. These audio smart glasses, unlike the others, have advantage in design. They look fashionable due to their small amount of electronics inside. Although I found a lot more AR Audio models, it is not necessary for this thesis to be all namely stated.

#### ***Frames by Bose***

Frames is a smart glasses model by Bose, which aims to replace headphones with glasses. The sound is transmitted through *bone induction* instead of putting buds inside ears. According to 100BuyTech, this model provides the best audio experience in terms of bone-conduction sound output nowadays. (100BuyTech, 2020)

Features: (Bose)

- Making phone calls
- Listening to music

- Notifications from phone
- Voice control
- Head gestures

### ***Gentle Monster by Huawei***

Gentle Monster by Huawei is audio smart glasses. They provide very similar features. Unlike Bose's Frames, they use directional speakers instead of bone-conduction, which is far more common in other smart glasses models. The unique feature is Monster's design. These are the only fashion-centered glasses in the list. (Huawei)

### ***Echo Frames by Amazon***

Another audio-only smart glasses model is made by Amazon. In addition to the previous model, they can read out notifications from your Android phone. *"You can also ask Alexa for rundowns of your calendar, the news, weather, and the usual things you've come to expect. A "VIP" filter will let wearers choose which notifications they want read aloud and which should remain only on their phone for later."* (Seifert, 2019) Features are similar to both models above, (Amazon) but they seem more concerned about everyday support than listening to music.

### ***Argon by Solos***

After Kopin Glasses, Solos aims to deliver other in-lens AR smart glasses, that focus on everyday life. This product is still in development but it runs a Kickstarter campaign. Its unique feature is interchangeable frames. They will also support health tracking, audio entertainment and personal assistance through AirGo mobile app. While it promises also "visual capabilities", the chances are that the product will be only audio based. There are no more details about those "visual capabilities" and it is hard to imagine interchangeable frames with built-in display. (Solos)

### 1.2.1.2 Audiovisual Models

Visual reality's perception can be enhanced by two kinds of displays – temple-mounted display (TMD) or in-lens display (ILD). I used this terminology because there was not any suitable term in the industry. Head-mounted display seemed a useful term at first sight but people tend to use it for any kind of display in front of eyes.

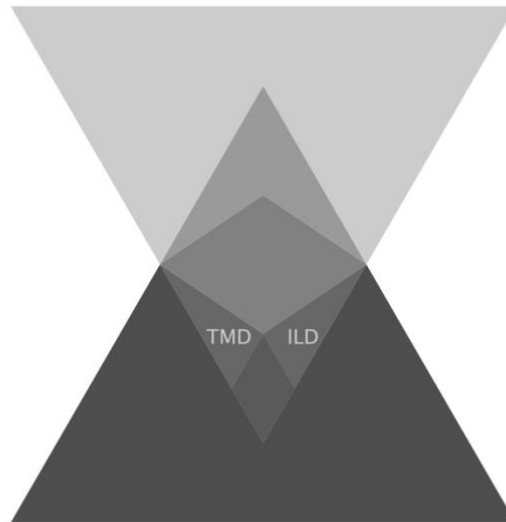


Figure 16 (Poláček, 2020)

#### 1.2.1.2.1 Temple-Mounted Display

In 2013 Google announced its Glass Explorer Edition that was supposed to make revolution. Undesirable publicity caused by privacy, health and design issues, stopped Google from mass distribution and slowly led to decline of this pioneer product. (Pohl, 2018)

This section contains models, that are very similar to Glasses by Google. They are monocular, made of a small display, clipped on a glasses' temple. This construction is too far away from what people would wear in public. This aspect moves all these eyewear devices (generally labeled smart glasses) in this section to professional sector, where professionals need them to improve their work efficiency. They do not care about what they look like.

In future, for these models might find their niche in sport, healthcare, engineering, manufacturing, logistics, security, military and other specialized sectors.

### ***Glasses by Google***

In 2017 Google announced return of the glasses in product line called Enterprise Edition, which moved the glasses from public to professional sector. Current latest model is Glass Enterprise Edition 2. However, this section contains description of the original Explorer Edition from 2013 model. The reasons are two. The first is only information about enhancements of the previous models in sources about Enterprise Edition 2. The second is that it is more valuable for this thesis to focus on Explorer Edition because it is made for publicity unlike Enterprise Edition. (Pohl, 2018)

Controls: (Trent)

- Voice input
- Camera button
  - Take a photo: Press
  - Take a video: Hold
- On-Head Detection
- Touchpad on right temple
  - Navigation among the screens you see in the display. Swipe forward and backward to see screens timeline.
  - Cancel an action by swiping from the top of the touchpad to the bottom
  - Tap to wake up
- Power button on the inner right side
- Diode on the back of the touchpad
  - Power status (white light blinks slowly while you're charging)

Google also provides guide for developers how to design apps for Glass (Google) and there are available examples (Wilson, 2014) (Phandroid, 2013) of how the Glass OS works.

### Features of Glasses by Google

### ***ORA by Optinvent***

ORA is one of the first successors of Google Glass, but less expensive ( Mobilegeeks.de, 2013). They introduced two modes of view. In Glance mode you see content below your field of vision, in AR mode the content is straight in your view. The current model is ORA-2 is oriented to the professional sector as Google did.

### ***dynaEdge by Toshiba***

Toshiba glasses, unlike the others, run Windows OS and not Android OS. They aim at professional market too. However, it is wired to power bank and controller in one. It is a standalone device that does not need user's phone to work. (Engadget, 2018)

### ***Glass C220 by Lenovo***

Lenovo runs several XR devices development at once. Its device Glass C220 is aimed at professionals too and wants to offer remote support functions by Augmented Human Cloud platform. (Palladino, 2017)

### ***M-Series by Vuzix***

Vuzix's M-Series aims at professionals in industrial sector. Their latest model is M400. Its features are Bluetooth, Wi-Fi, camera, Android OS, USB-C, sensors (gyroscope, accelerometer and magnetometer), speaker, microphone, GPS and notification diode. The glasses can be controlled by voice, touchpad (with multitouch support) or four buttons. (Vuzix) It also has Android-based UI. (Reality Forward, 2019)

### ***Varia Vision by Garmin***

Varia Vision by Garmin is monocular AR glasses for sport. It comes with vertically adjustable display (as it has ORA by Optinvent). They have two unique features.

The first is possibility to mount them on any glasses' temple – left or right on any frames or regular glasses. Due to this feature, the glasses become disbalanced.

Second is that they can be paired with other Garmin devices with specific sensors. Then on the display, the glasses can show from 1 to 4 sections containing the sensors' outputs. The less sections the display shows, the more information the in the sections are. The system is pretty simple anyway. (Halper, 2018)

### ***Recon Jet by Intel***

Recon Jet by Intel is also a sport-centered device. The unique feature is its low on-glass position. The glasses' computer is clipped on bottom of right lens, not on any temple as the rest of Temple-mounted models. The UI is very similar to Garmin's model. It has touch control (up, down + double tapping) and more complex menu. (Halper, 2018)

### ***Kopin by Solos***

Kopin by Solos is a sport-based smart glasses model. The glasses unique feature is micro display design, and possibility to chat with other Kopin's owners. (Halper, 2018) The UI is very similar to Garmin's model too. I also can be connected to sensors, phone (via app) and has vertically adjustable display. It can also be controlled by three buttons (up, down and enter) or by voice. It has no touch pad. The glasses have been founded through Kickstarter campaign. (Solos)

#### **1.2.1.2.2 In-Lens Display**

This sort of glasses differs from the previous one by design and display type. Here is the display part of the glasses' lens. Information is projected to inner side of lens, usually from right temple. The biggest advantage is their look, that finally might be accepted by public.

##### ***DK-52 by Lumus***

Lumus is Temple-mounted displays manufacturer, but it sells also its in-lens smart glasses called Development Kit (DK) 52. It is binocular AR glasses, that are thanks to it able to show 3D content.

While Lumus is basically a hardware provider, it sells its hardware to other businesses. They make their own products based on Lumus' hardware. (Lumus, 2019) (Fuller, 2017)

##### ***Blade by Vuzix***

Blade is another Vuzix Android-based smart glasses product. Although it is still industrial-centered, it has in-lens built-in monocular display. The developer provides great design guidelines (Vuzix) and quick start guide (Vuzix).

Controls: (Vuzix)

- Voice control
- Touch pad (multitouch support)
- Power button and 3 standard buttons
- Remote control app

##### ***Raptor by Every sight***

Raptor are the only in-lens display smart glasses for sport. It is a standalone unit with a camera. The manufacturer provides two user manuals, describing product features and controls (Every sight) (Every sight).

Controls:

- Touchpad
- Voice
- Third party Bluetooth devices (even mobile phone)

The glasses have a unique display, that shows 5 information at the same time, in the center, there can be a detail of navigation (what user can select by sliding forwards or backwards on the touchpad). (Halper, 2018)

### ***Focals by North***

Focals by North are smart glasses, that come with special ring controller with joystick instead of on-glasses touchpad. The developer is supported by Amazon (Isaak, 2019). North provides a great manual, explaining glasses features (North) and also companion app possibilities (North).

Features of Focals by North

### ***Norm by Human Capable***

Norm by Human Capable is a personal smart glasses model, still in development and was financed through Kickstarter (Human Capable, 2019) and IndieGoGo (Human Capable). The GUI has similar navigation to Focals by North. Their unique feature is their design and special display. The glasses can be controlled by voice, touchpad or head gestures. (Human Capable)

Features of Norm by Human Capable

## 1.2.2 Mixed Reality Glasses (headsets)

Previous smart glasses models lack one crucial feature – sufficient objects and surrounding tracking. Therefore, these models look more like headsets than glasses. Also, current MR headsets have more features suited for developers. Augmented Reality Glasses had more end-customer-ready UI.

### ***Moverio by Epson***

Moverio is binocular smart glasses series by Epson. All the models are paired to a remote or wired controller, so they have no on-glasses controls. There is a complete list of all current Moverio models: (Epson)

- Mobile extensions
  - BT-30C (lite device)
  - BT-35E (customized for commercial use)
- Standalone devices
  - BT-350 (customized for commercial use)
  - BT-300 (customized for personal use)
  - BT-2200 and BT-2000 (customized for industrial workplace)

For purposes of this thesis, there are considerable two models with user guides – BT-35E (Epson) and BT-300 (Epson). Although the glasses lack depth sensors, they offer a lot of MR possibilities thanks to camera tracking (Android Authority, 2015). The GUI of all these glasses is the same as Android tablets have (Captain Drone, 2018) (Ricker, 2017). It is not customized for wearable cases.

### ***Light by Nreal***

Light by Nreal are smart glasses, that are designed to be lightweight and comfortable. That is the reason why they are wired to a phone, that can take over the necessary hardware. Soon, the phone, eyes and hands can be used as a controller (Horwitz, 2020) (Heaney, 2020) together with new 3D form of its OS called Nebula (Ochanji, 2020). Currently the glasses have transparent UI if android-tablet look, placed in space (Ochanji, 2020) and are controlled by the round wireless controller (Nreal). The manufacturer also provides text-based design guidelines for developers (Nreal).

#### Features of Light by Nreal

### ***Glow by MAD Gaze***

Glow's predecessor is Vader glasses, that are MAD Gaze's first attempt to enter MR world. The GUI is classic Android tablet in front of your eyes (MAD Gaze, 2018).

Controls: (MAD Gaze)

- Buttons

- Hand gestures
- Voice
- Touchpad
- Remote app

The following model called Glow is based on Vader but has light fashionable construction. It is wired to user's phone and its unique feature is the infrared camera and LED diode indicator. The current latest model is Glow Plus, which is a slightly upgraded version of basic Glow model. (MAD Gaze)

### ***DK-Vision by Lumus***

Development Kit (DK) Vision (Lumus) by Lumus is its MR semi-finished product that has been used as a basement for ThinkReality by Lenovo (Lumus, 2019).

### ***ThinkReality by Lenovo***

Current ThinkReality model A6 by Lenovo is its MR model for enterprises (Finnegan, 2019). It uses ThinkReality Platform UI inside the headset (Lenovo).

### ***Glasses by Rokid***

Currently, Rokid works on two types of MR glasses – Glass 2 (Rokid) and Vision, a mobile extension (Rokid). The glasses are operated by Halo XUI system (AugmentedWorldExpo, 2019).

Glass 2 are wired to power bank and computer together and can be controlled by: (Usman, 2020) (AWE - Augmented World Expo, 2018)

- Voice
- Head gestures
- Touch

Features of Glasses by Rokid

### ***X2 by ThirdEye Gen***

X2 is ThirdEye Gen's smart glasses model for industrial purposes (ThirdEye Gen). Although they provide a broad user guide (ThirdEye Gen), the Gaze UI is a plain Android system without any special wearable customization (ThirdEye Gen, 2019).

### ***HoloLens by Microsoft***

Microsoft's current model HoloLens 2 is a standalone MR headset (Microsoft). Microsoft provides very good design development guide to its Windows for XR devices (Microsoft), which is partially based on Zero UI model, and shares learnings that they went through

(Microsoft). Some developers also wrote articles about design process for HoloLens (Marek, 2018) (UCL Peach Reality).

Features of HoloLens by Microsoft

### ***One by Magic Leap***

Magic Leap One is the company's standalone MR headset. It has a support by Google (Weinberger, 2015). The device is wired to an additional computer called Lightpack, that a user can put in his pocket. You can control the headset with a controller or your hands. (Magic Leap) It is powered by very intuitive Lumin OS (Torres, 2018), for which the manufacturer prepared advanced design guides (Magic Leap) and bloggers processed them (Dave Can Code, 2018) (Dave Can Code., 2018). It is likely that Magic Leap hired a design agency to design the whole UI for them (Threesixty Reality) (Argo Design).

Features of One by Magic Leap

### 1.2.3 Hypothetical Smart Glasses

This section contains smart glasses models, that haven't been officially announced yet, but there are reasons to believe that they will appear soon. Due to these companies' secretive culture, we must take this information with high level of uncertainty.

#### ***Glasses by Apple***

In 2007 Apple introduced first iPhone that ignites smartphone revolution (Apple, 2007). That is why general public might believe, that Apple might do it again with smart glasses. According to Tech Radar (Lynch, 2019), there might be wearable XR project called T288 at Apple, that might develop two models of eyewear devices. A headset and smart glasses that might also have, cameras with advanced human detection.

The AV headset would be used for maps, massaging, gaming, watching video and virtual meetings. It should have a surrounding scanning technology, an 8K display and an additional computer which it will connect with headset via a high-speed, short-range wireless technology. (Painter, 2020)

The smart glasses should have a bone conduction earpiece, a microphone, an accelerometer and a capacitive frame for navigation and interaction. The lenses should get darker on bright sun, have some kind of opacity system. They should be also connected to iPhone that will process all the data. (Painter, 2020)

#### ***Glasses by Facebook***

According to CNBC (Rodriguez, 2017), Facebook might also work on its version of AR smart glasses in project called Orion. *“The glasses would allow users to take calls, show information to users in a small display and live-stream their vantage point to their social media friends and followers. Facebook is also developing an artificial intelligence voice assistant that would serve as a user input for the glasses. In addition, the company has experimented with a ring device that would allow users to input information via motion sensor. That device is code-named Agios.”*

#### ***Glasses by Lenovo***

Alongside other smart glasses models, Lenovo might work also on another eyewear device, but in this case the model is suited for business travelers (Hayden, 2019) (Lenovo, 2019).

## 1.2.4 Discontinued Smart Glasses

### ***Vaunt by Intel***

Vaunt was a prototype of smart glasses developed by Intel. It had an in-lens display, that was able to project information in monochromatic red tones (The Verge, 2018). However, the project was shut down in 2018, because of Intel's inability to deliver the product to the customers directly. (Bohn, 2018)

### ***R-7 by ODG***

ODG was a pioneer smart glasses company with long tradition and big support from large enterprises. Their latest model R-7 was professional AR smart glasses with an in-lens display, showing Android tablet UI. The source of the company's 2018 collapse was financial problems, caused by focus on too wide products focus. (Matney, 2019)

### ***SmartEyeglass SED-E1 by Sony***

SmartEyeglass was an AR smart glasses model by Sony from 2015, released for developers. It has in-lens display showing a simple green monochrome text and diagrams. It was able to connect to Android smartphone and was controlled by a wired controller with buttons and a touchpad. It had a camera, a speaker, a microphone and NFC (Lawler, 2015). The model was shut down before getting to customers (Sony). It still provides a user guide though (Sony, 2017).

### ***Glasses by Meta***

Meta was a pioneer MR smart glasses developer (Meta, 2013). Its latest model Meta 2 was controlled by hand gestures, was wired to an extra computer and had its own OS based on Zero UI (Piltch, 2014). The company was sold in 2017 due to financial problems, caused by USA's trade war with China. However, the buyer promised bright future for Meta's hardware and software (Robertso, 2018).

## 1.2.5 Augmented Virtuality Headsets

Some credible sources (Cherdo, 2020) (Jaehnig, 2020) include AV headsets under smart glasses. But these devices can never become smart glasses by our definition. The reason is, that AV is a superstructure of VR. And VR aims to deliver out-of-world experience, without distractions from real world. So there is no reason to open up headsets and start evolving them to glasses, which kills virtuality.

There is a short list of significant AV devices:

- Windows Mixed Reality Headset by Asus
- XR-1 by Varjo
- Mirage by Lenovo

## 1.2.6 Summary of Smart Glasses Models

This selection contains a table comparison, made from Smart Glasses Models list above, which takes into accounts these conditions for the further UI concept:

- A. **Everyday Usability** (preferred are models, that are developed for end-point customer, who uses the device on everyday basis).
- B. **Progressivity** (preferred are technologically advanced models, that offers intensive audiovisual experience).

The values below indicate, how much the selected model meets the specific condition in percentages. The result is counted as a multiple of percentages from both conditions. I chode to prefer (as an inspiration) those models, that have result greater than 50% included.

*Note: Lumus models were dismissed, because Lumus provides only hardware, not market-ready products.*

Table 1 (Poláček, 2020)v

| No | Name                            | Everyday Usability | Progressivity | Result |
|----|---------------------------------|--------------------|---------------|--------|
| 1  | <u>Spectacles by Snap</u>       | 100%               | 25%           | 25%    |
| 2  | <u>Frames by Bose</u>           | 100%               | 25%           | 25%    |
| 3  | <u>Gentle Monster by Huawei</u> | 100%               | 25%           | 25%    |
| 4  | <u>Echo Frames by Amazon</u>    | 100%               | 25%           | 25%    |
| 5  | <u>Argon by Solos</u>           | 100%               | 25%           | 25%    |
| 6  | <u>Glasses by Google</u>        | 100%               | 50%           | 50%    |
| 7  | <u>ORA by Optinvent</u>         | 25%                | 50%           | 13%    |
| 8  | <u>dynaEdge by Toshiba</u>      | 25%                | 50%           | 13%    |
| 9  | <u>Glass C220 by Lenovo</u>     | 25%                | 50%           | 13%    |
| 10 | <u>M-Series by Vuzix</u>        | 25%                | 50%           | 13%    |
| 11 | <u>Varia Vision by Garmin</u>   | 50%                | 50%           | 25%    |
| 12 | <u>Recon Jet by Intel</u>       | 50%                | 50%           | 25%    |
| 13 | <u>Kopin by Solos</u>           | 50%                | 50%           | 25%    |

|    |                              |      |      |      |
|----|------------------------------|------|------|------|
| 14 | Blade by Vuzix               | 50%  | 75%  | 38%  |
| 15 | Raptor by EverySight         | 50%  | 75%  | 38%  |
| 16 | Focals by North              | 100% | 75%  | 75%  |
| 17 | Norm by Human Capable        | 100% | 75%  | 75%  |
| 18 | Moverio by Epson             | 50%  | 75%  | 38%  |
| 19 | Light by Nreal               | 100% | 100% | 100% |
| 20 | Glow by MAD Gaze             | 100% | 100% | 100% |
| 21 | ThinkReality by Lenovo       | 25%  | 100% | 25%  |
| 22 | Glasses by Rokid             | 100% | 100% | 100% |
| 23 | X2 by ThirdEye Gen           | 25%  | 100% | 25%  |
| 24 | HoloLens by Microsoft        | 50%  | 100% | 50%  |
| 25 | One by Magic Leap            | 75%  | 100% | 75%  |
| 26 | Glasses by Apple             | ?    | ?    | ?    |
| 27 | Glasses by Facebook          | ?    | ?    | ?    |
| 28 | Glasses by Lenovo            | ?    | ?    | ?    |
| 29 | Vaunt by Intel               | x    | 75%  | x    |
| 30 | R-7 by ODG                   | x    | 75%  | x    |
| 31 | SmartEyeglass SED-E1 by Sony | x    | 75%  | x    |
| 32 | Glasses by Meta              | x    | 100% | x    |

Also, it is convenient to follow developers, who:

- Have probability that their model will become mainstream and standard for other concepts designed in future. So, in the results, there can be prioritized developers with competitive advantage.
- Have good resources that can be used as knowledge basis for the UI concept.

The selected models are:

- One by Magic Leap (offers design guide, backed by Google)
- HoloLens by Microsoft (offers design guide, strong developer)
- Glasses by Google (offers design guide, strong developer)
- Light by Nreal (offers design guide)
- Focals by North (backed by Amazon)
- Glasses by Rokid
- Glow by MAD Gaze
- Norm by Human Capable

It might be valuable for future-centered UI concept to take into account also hypothetical developers' plans, because they have potential to deliver customer-ready smart glasses with massive impact.

- Glasses by Apple
- Glasses by Facebook
- Glasses by Lenovo

## **1.3 Preconditions for the UI Design**

After the market research, there is a need to analyze methods, used in UI design processes for eyewear devices.

## 1.3.1 View Possibilities of Content

Due to operations within a 3D environment, not in a screen, there are different ways how the content inside the glasses can appear.

### 1.3.1.1 Scene Type

Scene type defines, how certain app is visualized. The app can look as a standard screen (as we know from desktop) or it can be placed on objects around you in your surroundings.

#### ***Landscape***

Landscape apps display 2D content. They are virtual screens, that we usually look at when we are working on our laptops or mobiles.

#### ***Immersive***

Immersive apps display 3D content. The content usually blends with our surrounding, providing view of true mixed reality.

### 1.3.1.2 Position of the Scene

Position of the scene defines, where certain app is placed in space. The app can be stucked to you wherever you go or it can stay pinned on certain location.

#### ***Absolute***

Absolute view commands scene to stay in front of our view, no matter where we are located or where we go. When we look away from the scene it is still in front of our eyes.

#### ***Relative***

Relative view commands scene to be located at a certain anchor point in our surrounding. When we look away from the pinned scene we cannot see it as in reality.

### 1.3.1.3 View Zones (1/2)

View zones show, where digital content should appear to be properly registered by user's eyes. Placement of the information is crucial, because it defines its importance, relevance and visibility.

#### 1.3.1.3.1 2D Zones (1/2)

This section deals with vertical and horizontal placement of a piece of information.

## 2D View Zones (1/2) – Experiment

*I decided to run an experiment, because I was not able to find proper sources about information structure inside smart glasses' 2D view. As this section contains results based on cursory experiment, please keep in mind that the results might not be accurate. The discipline of ocular vision is extremely vast, and this thesis cannot cover fully this area.*

For purposes of this experiment, I developed a prototype of smart glasses with lenses replaced by transparent foil. This experimenting showed, how to design 2D content for smart glasses.



Figure 17 (Poláček, 2020)

For recording of the results I used free stock image (RoyalHenrey, 2017).

### The experiment results:

1. Eyes have areas, where they perceive intensively (A), less (B) and where almost not (B). The latter ones are called peripheral areas.

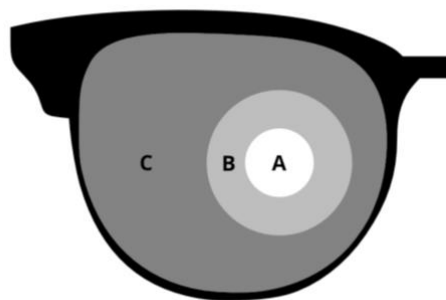


Figure 18 (Poláček, 2020)

2. There is a border in our view, while looking to left and right side. It is caused by inner circle of the frames around eyes on nose side. The visible inner frame worsens visibility of content that would be placed there.



Figure 19 (Poláček, 2020)

3. Another forbidden area (areas) are inner halves of the lenses next to the nose. These areas blend together while looking through glasses.



Figure 20 (Poláček, 2020)

4. Eyes tend to look in wide view (Media College). That might be the reason of medieval helmet peepholes wide design.



Figure 21 (Poláček, 2020)

Here I came with model, where the lenses are partitioned, and individual areas are separated.



Figure 22 (Poláček, 2020)

There I marked the areas from Figure 22 and studied, how respondents react to them. I chose only right eye lens, because of ocular dominance. 70% of people are right-eye-dominant and only 30% is left-eye-dominant (Halper, 2018), so I focused on the majority of people. The aim was to determine, which of the marked areas on right lens in periphery area (area marked C in Figure 18) is the least intrusive and at the same time the most accessible to look at (1 is best, 2 is worst).

Respondent 1 (female, 49 years, high school teacher)

1. Top right
2. Bottom left

Respondent 2 (female, 20 years, university student)

1. Top right
2. Bottom left

Respondent 3 (male, 23 years, university student)

1. Top left
2. Bottom right

Respondent 4 (female, 11 years, primary school student)

1. Top right
2. Bottom left

Respondent 5 (male, 49 years, quality manager)

1. Top right
2. Bottom left

The results showed, that the top right placement of temple-mounted displays (like Glasses by Google and other similar models) is good option to display occasional content to glance at.

The result from Respondent 3 is probably only an exception, because all the respondents were right-handed by later findings.

#### 1.3.1.3.2 3D Zones (1/2)

This section deals with object's distance from the user. There are a few zones around (Blokša, 2017):

1. **No-No Zone:** right around the head where nothing should be permanently (0,5 meters from eyes)
2. **Content Zone:** around the user from 0,5 to 20 meters
  - a. The Main Content Zone: ahead of the user from left to right. To interact with hands the content needs to be from 0,5 to 1 meter from user (authors' note)
    - i. Ideal Landscape app placement zone to only watch is 2 meters
    - ii. Ideal Immersive app placement zone is from 1,25 to 5 meters from the user
  - b. The Peripheral Zone: on sides of the user, where the user still can see, but it is not meant for necessary elements
  - c. Curiosity Zone: back behind the user, where the user has to turn around to see the content

## 1.3.2 Adaptability Between Devices

This section consists of UX transfer and UI evolution, heading towards eyewear devices. The evolution shows, that the interactions between user and device are quicker and more frequent.

Smart glasses might start as an extension for smart phones and then become a separate unit. Just like once laptop became extension of PC, mobile phone became extension of laptop and smart watches are now extension of smart phones. *“The thing is that the extension device, once matured, usually becomes independent to its base station.”* (Kopta, 2020)

Assuming this experience, there has to be smooth transition between these devices' UIs in order to maintain comfortable environment and to avoid learning difficulties. These factors might ensure better acceptance by consumers.

### 1.3.2.1 Desktops

The major OS on desktop devices are currently **Windows** by Microsoft (77% of market share) and **macOS** by Apple (18% of market share). (StatCounter, 2020) Being too far away from smart glasses, wearable devices inherited only few nowadays UI desktop components. Those are:

- Customizable desktop (home screen)
- Ribbon info center
- Programs (apps) pinning
- All programs (apps) list
- Point and Click UI (using mouse and keyboard)

### 1.3.2.2 Smart Phones

First, mobile phones had a few main features. After releasing iPhone 2G by Apple, mobiles got touch screens, programs turned into apps, the smartphone revolution started and now we experience its peak.

The major OS on mobile devices are **Android** by Google (72%) and **iOS** by Apple (27%). (StatsCount, 2020) There, wearable devices inherited a lot more UI components. Those are:

- Notifications
- Separated Control center and Settings
- Lock screen
- Widgets
- Apps timeline (history)
- Touch Gestures

### 1.3.2.3 Smart Watches

Smart watches are nowadays good examples of functional mainstream wearable devices. The major OS on mobile devices are **watchOS** by Apple (41%), **Wear OS** (formerly Android Wear) by Google (7%), and **Tizen** by Samsung (2%). (Statista, 2018)

*„The UI of smart watches is based on mobile phones, with some differences: The small display area forced the manufacturers to let the user perform gestures on on-device hardware controls (crown, display surface, buttons) instead of directly manipulating the UI element (typical for Natural UI in mobile phones and tablets).“* (Kopta, 2020)

#### **watchOS**

Unlike other smart watches' manufacturers, Apple fits its watchOS on rectangular display, instead of traditional circular. Apple also provides design guidelines for apps' developers (Apple).

Here is the GUI, its controls and default apps: (AppFind, 2019) (Apple)

#### **GUI Screens**

- Face (home screen)
  - Force touch: Customize Face
  - Press Digital Crown: Apps
  - Swipe down: Notifications
  - Swipe up: Glances
    - Swipe right: Control Center
    - Swipe left: Widgets

#### **Default apps**

- Activity
  - Goals
  - Workout
- Weather
- Maps
- Music
- Mail
- World Clock
- Alarm
- Settings
- Photos
- Stopwatch
- Timer
- Calling
- Messages
- Remote trigger

- Calendar
- Passbook
- Stocks

## ***Wear OS***

Wear OS, formerly Android Wear, is another OS that powers smart watches of foreign manufacturers. Like Apple, Google also provides design guidelines for developers (Google, 2019) (Google).

Here is the GUI and its controls: (AppFind, 2019) (Fossil, 2018)

### **GUI Screens**

- Face (home screen)
  - Swipe down: Control Center
  - Swipe up: Notifications
  - Swipe right: Suggestions
  - Swipe left: Widgets

## ***Tizen***

Samsung usually uses Android in its phones, but in case of watches it used its own OS called Tizen.

Here is the GUI and its controls: (AppFind, 2019) (Samsung)

### **GUI Screens**

- Face (home screen)
  - Swipe down: Control Center
  - Swipe right: Notifications
  - Swipe left: Widgets

## **1.3.2.4 Smart Glasses**

### ***Zero UI***

In recent years, while technology started becoming more advanced and with arrival of new devices, new interfaces start to emerge (e.g. haptic interfaces, ambient interfaces). There is one, in my opinion, very suited for eyewear XR devices called Zero UI or No UI, that uses human body as a controller. HoloLens by Microsoft or discontinued model Glasses by Meta did very good job in terms of Zero UI.

Now we are so used to GUI that natural UI like Zero UI might seems far exotic. We need to re-learn people to interact naturally with digital products by slow replacement of GUI components with Zero UI.

### 1.3.3 Universal Feature Set Definition

Determining which features should smartwatch UI contain is necessary to persuade user to use smart glasses.

#### 1.3.3.1 General Audience Needs Research

This section contains features, that general audience needs every day at different smart devices.

##### 1.3.3.1.1 Smartphone User Needs

Very good device for doing everyday use cases research is smartphone. There is huge mass of respondents, but not all the features are transferable to glasses (e.g. web browsing).

#### *Secondary Research*

Rank from Axway (Axway, 2017):

1. Messaging - 70 %
2. Phone calls - 62 %
3. Email - 54 %
4. Social networking - 51 %
5. Camera - 37 %
6. Gaming - 30 %
7. Music - 28 %
8. Directions/Maps - 24 %
9. Clock - 23 %
10. Shopping - 23 %
11. Weather - 22 %
12. Banking - 21 %
13. Mobile TV - 11 %
14. Video chat - 8 %

Rank from Gadget Cover (Brown, 2019):

1. Text - 88 %
2. Email - 70 %
3. Facebook - 62 %
4. Camera - 61 %
5. Reading news - 58 %
6. Online shopping - 56 %
7. Checking the weather - 54 %
8. WhatsApp - 51 %
9. Banking - 45 %
10. Watching videos on YouTube - 42 %

### **Primary Research**

This research was carried out by face-to-face short survey among representatives of different ages, genders, professions and hobbies from a mid-size town (of 15 000 citizens) in the western Czech Republic. The representatives were asked to name up to 10 most used smart phone apps or features on everyday basis. The aim was to confirm or reject the results of secondary research (from above) in local area (the Czech Republic).

Respondent 1 (female, 49 years, high school teacher)

1. Calling and SMS
2. Photos and Camera
3. Facebook
4. WhatsApp
5. Instagram
6. Searching
7. Banking
8. Professional apps

Respondent 2 (female, 20 years, university student)

1. Messenger
2. Instagram
3. YouTube
4. Mail
5. Camera
6. Maps
7. Facebook
8. Browser
9. Shopping
10. Activity Tracking

Respondent 3 (male, 23 years, university student)

1. Instagram, Twitter, Facebook
2. News
3. Maps
4. YouTube
5. Music
6. Texting (Email, WhatsApp, Messenger)
7. Notes
8. Shopping

Respondent 4 (female, 11 years, primary school student)

1. TikTok
2. Instagram
3. WhatsApp
4. Calling

5. Camera
6. Roblox
7. YouTube (music + videos)
8. Snapchat
9. Email
10. Time

Respondent 5 (male, 49 years, quality manager)

1. Calling
2. Reminders (voice, calendar)
3. Music
4. Camera
5. Texting (SMS, WhatsApp, Messenger)
6. Navigation
7. Emails
8. Facebook
9. News
10. Banking
11. Weather

My personal investigation has confirmed the web results.

#### **1.3.3.1.2 Smartwatch User Needs**

Based on my investigation so far, smart watches might be perceived as a bridge between smart phone and smart glasses.

#### ***Secondary Research***

Rank from Statista (Richter, 2017):

1. Notifications/text – 54 %
2. Activity tracking – 45 %
3. News updates – 31 %
4. Phone calls – 26 %
5. Alarm clock – 25 %
6. Email – 25 %
7. View photos/videos – 21 %
8. GPS/navigation – 19 %
9. Remote control music – 18 %
10. Home automation – 17 %

#### ***Primary Research***

This research was carried out by short online survey among representatives of different genders, professions and hobbies from all over the Czech Republic, who use smart watches.

The representatives were asked to name up to 10 most used smart watch apps or features on everyday basis. The aim was to confirm or reject the results of secondary research (from above) in local area (the Czech Republic).

Respondent 1 (male, 20 years, university student)

1. Activity Tracking
2. Notifications
3. Calling
4. Texting
5. Music
6. Apple Pay

Respondent 2 (male, 18 years, high school student)

1. Time
2. Notifications
3. Texting
4. Apple Pay
5. Music
6. Calling
7. Activity Tracking
8. Alarm
9. Navigation

Respondent 3 (male, 20 years, university student)

1. Messages
2. Time
3. Notifications
4. Design
5. Music

Respondent 4 (male, 22 years, university student)

1. Photos and Camera
2. Music
3. Name days App + Faces
4. Activity Tracking
5. Apple Pay
6. Design
7. Remote Control
8. Notifications
9. Weather

Respondent 5 (female, 22 years, university student)

1. Activity Tracking
2. Remote Control

3. Time
4. Design

### 1.3.3.1.3 Smart Glasses User Needs

Due to missing existence of smart glasses on mainstream market, we can only estimate what people might use on daily basis.

Steve Jobs once said:

*“Some people say, ‘Give the customers what they want.’ But that’s not my approach. Our job is to figure out what they’re going to want before they do. I think Henry Ford once said, “If I’d asked customers what they wanted, they would have told me, ‘A faster horse!’” People don’t know what they want until you show it to them. That’s why I never rely on market research. Our task is to read things that are not yet on the page.”* (Smith, 2019)

### **Most Popular AR & MR Apps**

Today’s AR and MR apps are based usually on entertainment. They are not focused on usable features for which we would buy and use the device (as we started buying mobiles when we needed to call and text from somewhere else than from home). Current successful and together useful apps on the market are: (Corpuz, 2020) (Dube, 2018) (Jansen, 2019) (Sheehan, 2019)

- Measurements – AR Ruler
- Try-on apps
  - Makeup – YouCam MakeUp
  - Hair – Snapchat
  - Clothes & Fashion – Amikasa; Wanna Kicks
  - Tattoo – Inkhunter
  - Furniture – Ikea Place; Houzz
- Maps
  - Navigation – Google Maps
  - Nearby places viewer – W.A.R.; ViewRanger
  - Car directions – Augmented Car Finder
  - Safe places – Spot Crime
  - Compass – Augmented Reality Compass and AR Compass 3D
- Visual search – Google Lens
- Sky mapper
  - Night – Star Walk 2; Big Bang AR; Sky View
  - Day – Sun Seeker
- Tagging World – WallaMe
- Remote assistance – Vuforia Chalk
- Visual Translate – Google Translate
- Drawing support (image tracing) – Sketch AR
- Calculator – MyScript Calculator 2 (Hindy, 2019)

- Magnifier/Telescope (camera feature)

### 1.3.3.2 Smart Glasses Use Cases

First, computers were only for calculations but they evolved to multifunctional devices. Phones were only used for calling and texting. But they evolved to multifunctional device, overtaking computers as personal devices (not professional working devices). Smartwatches have been here since 2014 (Apple, 2014) and they do not seem to overtake smart phones as personal devices. They seem to improve things that smart phones can never improve because of their construction.

Smart glasses come to the market with a set of features, that the manufacturers believe are good for their customers. They can be taken also as a result of their primary research, that we can use (for purpose of this thesis, this list includes only models from the section Summary of Smart Glasses Models).

#### ***Features of One by Magic Leap***

(Magic Leap, 2020)

- Gallery
- World (app store)
- Helio (internet browser)
- Screens (multiple virtual desktops)
- Social (communicator with other Magic Leap owners)
- Create (2D drawing studio and 3D animated objects gallery)
- All Apps
- Others
- User Guide
- Coming Soon

#### ***Features of HoloLens by Microsoft***

(Bright, 2016)

- Photos
- Holograms (3D animated objects gallery)
- Edge (browser)
- 3D Viewer (3D objects editor)
- Mail
- Calibration
- Cortana (voice assistant)
- Skype
- Store
- Settings

- Learn Gestures
- Get started
- Feedback

### ***Features of Glasses by Rokid***

(Rokid)

- Camera
- Gallery
- Face ID
- Remote
- Settings
- Document

### ***Features of Light by Nreal***

(Ochanji, 2020)

I was not able to find proper sources about any of Light's apps, so the purpose of them can be only estimated.

- Gallery
- Infinity Screen
- Infinity Space
- Portal
- Nreal Insight
- Discover MR
- Smart house
- Kingdom of Blades
- AR Shoebox

### ***Features of Glow by MAD Gaze***

(MAD Gaze)

- Sports Companion
- Calls
- 3D movies
- Gaming
- Documents Editor
- Navigation
- Translation
- Messaging
- Live Streaming

### ***Features of Focals by North***

(North)

- Now (home screen – date and time, calendar, control center)
- Messages
- Calendar
- Places (navigation)
- Weather
- Amazon Alexa
- Notifications
- Speaker Notes (presentations companion)

### ***Features of Norm by Human Capable***

(Human Capable) (Human Capable)

- Home (showing time and day)
- Camera (photo, video)
- Gallery
- Contacts
- Calling, messaging
- Sensor
- Music, podcasts, audiobooks
- Voice Recorder
- Settings
- Weather
- Maps and navigation
- QR scanner
- Browser
- Video player
- Translating
- Reminders
- Email
- Social Media
- Live Streaming
- Notes

### ***Features of Glasses by Google***

(Brownlee, 2013)

- Take a picture
- Record a video
- Get directions

- Send a message
- Make a phone call
- Start a Google Hangouts conversation
- Search on Google

## 1.4 Visual Design (1/2)

*“Visual design is a polarizing piece of the design process. It's often ignored as a superfluous facet of design, or it is viewed as the end-goal of great design. Visual design, however, is neither insignificant nor the end-all to great design. Visual design is a means of communicating through orienting illustrations and photography, meaningful typography, evocative iconography, persuasive color, comforting spacing and layout execution, and other little big details.”* (Rhyne, 2016)

### **Elements Size**

Due to distance of the glasses' lenses from your eyes, the elements on the display has to be very small. Their size is visible for example from look inside the lenses of Focals by North (Fisher, 2019).

### **Feedback Problem**

As it comes to hand interaction with virtual objects, it is hard to feel a response and the action is not satisfactory. Digital objects have no substance, so we cannot touch them physically, which leaves us unsure, whether the action (e.g. pressing the button) was successful. (Blokša, 2017)

## 1.4.1 Current Models' Graphics

Based on the Smart Glasses Models research, there are two approaches among smart glasses' manufacturers in **Error! Reference source not found.**

### **AR Approach**

This approach is presented in Audiovisual Models with In-Lens Display smart glasses. It is slightly inconsistent among all the models. However, other GUIs follow similar approach. They contain simple outlined icons with white text and the content has only informative character. Neither background, nor virtual buttons are presented.

### **MR Approach**

This approach is presented in Mixed Reality Glasses (headsets) and is highly inconsistent among all the models. It means that there is still not a stable common format of GUI. But there are MR manufacturers who have eyewear-device-ready GUI:

- One by Magic Leap contains 3D icons in flat design style. Its OS Lumin does not acknowledge Landscape UI if not necessary. If there is a text-based screen, it consists of white text without any background. It also features special icons. The icon (or other element) selection is represented by appearance of a gradient background behind the icon, that fades-out to surrounding. (Magic Leap)
- HoloLens by Microsoft acknowledge Landscape UI and follows its windowed attitude. It uses usually screens with a colorful background. (Microsoft) Microsoft uses its new Fluent Design System. It is built with an idea, that following this design language allows all digital designs to work great on any device, across all platforms. (Microsoft)
- Glasses by Rokid combines colorful AR Approach from Focals by North together with MR Approach (transparent text-based screens and icons with gradient background) from One by Magic Leap. (UPHIGH Productions, 2019)
- ThinkReality by Lenovo has very plain GUI, customized for workers in manufacturing environment.
- Light by Nreal offers slightly customized Android-tablet GUI with transparency (Ochanji, 2020)

## 1.4.2 Directions of Digital Graphics

Before 1970's, there were not many digital tools to create a graphic content. Although, *“early imaging methods included embossing and engraving, which led to 3D effects that Western culture still wears (e.g. in serifs of writing), graphics was usually 2D”* (Kopta, 2020). Since 1970's, when 3D special effects started to appear in movies, 3D started to gradually appear in more and more graphic content. (Cheddar, 2019)

### ***Skeuomorphic Theory***

The real change came in early 2000s, when new graphic tools started to appear together with more efficient computers. The digital graphics started to turn to 3D and imitate real life objects (started to be skeuomorphic). This appearance allowed people to better navigate in digital UIs. Especially in touch screen devices, where elements looked more real. They encouraged people to touch them together with vibrations and clicking sounds. *“3D logo designs were vital to make people feel comfortable in the new developing age.”* (Cheddar, 2019)

### ***Flat Theory***

*“As the world got more comfortable using these technologies, 3D design became less important and graphic designers started to shift back towards 2D.”* Flat graphic design started to be popular in 2013, when Apple introduces iOS 7 together with new UI visual. The first feedback from public was negative, because the change was huge. Even so, soon after all the world's companies followed this design trend. Although the lack of plasticity left users confused of what is clickable and what is not, it enabled them to focus only on important elements. (Cheddar, 2019)



Figure 23 (Popicon)

## 2 Practical Part

Today, fashionable AR glasses, focusing on everyday use, are too weak to do MR. And MR headsets focus mostly on entertainment and development, not on practical everyday use. After I went through all of smart glasses industry I believe that combination of AR glasses everyday approach with MR headsets 3D possibilities is the future of smart glasses.

### ***In Common AR, MR and AV Term***

There is missing term, that would cover all forms of reality, where the real world connects to a virtual one. There are separated terms for AR, MR and AV but there should be something, that would cover all three of them. For example, term Hybrid Reality (which has not any exact meaning yet) might serve this purpose.

### ***Phone Extension***

As it has been mentioned in this thesis – phones will probably be smart glasses companion (or vice versa) for a long time. Therefore, a lot of features that are designed are based on the fact, that smartphone is always present in the user's pocket. For example, replying to a mail is better to handle on keyboard, so if there would be a Mail app in smart glasses, it would have just basic features (e.g. delete mail, quick response or possibly just forward).

## 2.1 Input Interface

Yet users have all been used to GUI from our phones or laptops. We got used to some input methods but new devices require new ones. However, the methods cannot be too new or innovative, because in such form they can confuse and scare off new users. Therefore, the “old” input still has to be partially maintained. (Ing. Zdeněk Vondra, 2017)

### ***Input Methods Combination***

Every use case requires a different input method. *“As there is a large scale of control types for AR, the designer has to take in mind whether the control type is suitable for the specific situation. A worker in a noisy environment is probably not going to use voice to control AR instruction manual. To strictly have only one type of control will limit AR device usage. Therefore, there should be an opportunity of control type change.”* (Blokša, 2017)

Every input type should be considered for every feature wisely. But the inputs can be also combined to achieve better results (e.g. using gestures and control devices to achieve more accuracy). (Blokša, 2017)

To sum it up, product should be controllable by different ways and not to force the user to use one way.

## 2.1.1 Private Controls

When users interact with devices, there is still cold approach towards “communication to nobody” input methods (e.g. voice commands). People have been using these means of communication when they were in contact with a real person for thousands of years. It might be the reason why it is so strange seeing people calling using their Air Pods without spotting a headphone wire or a mobile. This would be probably similar with in-air hand gestures or head gestures.

In privacy, people are not aware of others. For example, daily commands’ frequencies are usually 4 times higher when using a smart speaker at home than using a mobile (Kinsella, 2018). This supports the theory, that people use voice commands often when others cannot hear them.

### 2.1.1.1 Hand Gestures

In MR, it is necessary to handle virtual 3D objects in space. In real life we use our hands that have 6 DoF – rotations in 3 axis and shift in 3 axis (Barnard, 2019) – and fingers for fine movements. If there is a natural way of control, it should be maintained and not replaced by a controller device.

Around the world, there are a lot of cultures, that use the same gestures, but their meaning differs. So, standardization and overcoming these obstacles might become a challenge for designers in future (Bharadwaj, 2017).

### 2.1.1.2 Voice Commands

Virtual assistants like Siri, Alexa, Cortana, Watson, and others that rely on artificial intelligence are more natural interface for humans than a traditional visual UI of windows and buttons (Bharadwaj, 2017). They can perfectly replace a keyboard in the wearables’ world.

Yet, it is still strange to talk to “nobody” while we are in public. People usually do not use voice commands there (Halper, 2018). It also avoids our privacy. This is the reason that voice commands can never fully replace hand control.

### 2.1.1.3 Head Gestures

Head gestures use only Frames by Bose to control volume. However, they can be used to accept (confirm) or reject (cancel) some coming in information. Although head gestures are not popular among smart glasses manufacturers, head shaking is our natural way of communication so it should be used at least as an additional input method.

## 2.1.2 Public Controls

If people are in public, they want to keep their inputs private. They are also concerned about not being perceived as weird people, who talk and make gestures to emptiness.

### 2.1.2.1 Eye-Tracking

Eye-tracking works as a specific element selector. It is used mostly in combination with touchpad gestures, that open the selected element.

The selection tool is not available with eye-tracking in this UI concept. If it worked that way, Microsoft HoloLens, which use eye tracking (TWiT Tech Podcast Network, 2019), would use it.

### 2.1.2.2 On-Glasses Controls

On-glass controls are common interaction elements of current AR glasses models. Moreover, it is a perfect way of navigation in 2D UIs and people already got used to swiping gestures.

#### 2.1.2.2.1 Touchpad

Touchpad is able to provide a lot of inputs and we have been using it for decades on laptops. Multitouch and swiping gestures are well known from smart phones too. Touchpad gestures are performed at outer side of right temple, where touchpad is (see the [General Skeleton of Glasses](#)).

### Gestures

- One finger (often used features)
  - Swipe forward or backward: Scroll through content or elements
  - Swipe down: All apps
  - Tap: Open element (*Tap twice - Win / Press touchpad – macOS*)
  - Tap and hold: Edit element
- Two fingers (less used features)
  - Swipe forward or backward: ([Head Gestures](#) alternative)
    - Backward (“come here” gesture): accept (confirm) incoming information
    - Forward (“get away” gesture): reject (cancel) incoming information
  - Swipe down: Notifications
  - Tap: Voice assistant
  - Tap and hold: Volume control (swipe to adjust volume)
- Three fingers (complicated to use, the touchpad is not enough large)
  - Swipe forward or backward: Copy or Paste
  - Swipe down: (*customizable*)

- Tap: (*customizable*)
- Tap and hold: (*customizable*)

#### **2.1.2.2.2 Power Button**

Buttons have been here a lot longer, but their amount on devices is significantly decreasing to usually one (mostly power button), two to three buttons (power, volume and Silent Mode switch) nowadays.

#### **2.1.2.3 Control Device**

Although fingers are good input device, sometimes people become tired and do not want to even lift their arm. There should be an alternative for using smart glasses without lifting arm up.

#### ***Hardware Requirements***

The controller should have 6 DoF sensor to imitate a human hand for object control in 3D MR environment. It also needs to have touchpad for swiping inputs. This can be achieved fully with a separate controller. Or else the controller can be a smart phone (a lot of people use smart phones now and they might use them for a long time, so they might not get motivation to buy another device as a controller). Interestingly, Focals by North or Glasses by Facebook are controlled by a smart ring controller.

#### ***Backwards Compatibility***

There should also be maintained backwards compatibility with desktop and traditional controllers like a mouse and a keyboard. The control device can work as a mouse (called “laser pointer” in XR industry) and voice commands can work as a keyboard. Alternatively MR can imitate both by projecting a hologram of the mouse or the keyboard.

## 2.2 General Skeleton of Glasses

The following pictures are wireframes of an ideal smart glasses model for everyday personal use. It includes common physical features and controls, that personal models have according to my research.

### ***Front View***



Figure 24 (Poláček, 2020)

Created with free stock image by (RoyalHenrey, 2017)

Contains:

- Camera
- LED notification diode (indicator of running Camera app – privacy element)
- Eye tracking sensors
- Nose pads
- Flashlight
- Surrounding detector

***Top View***

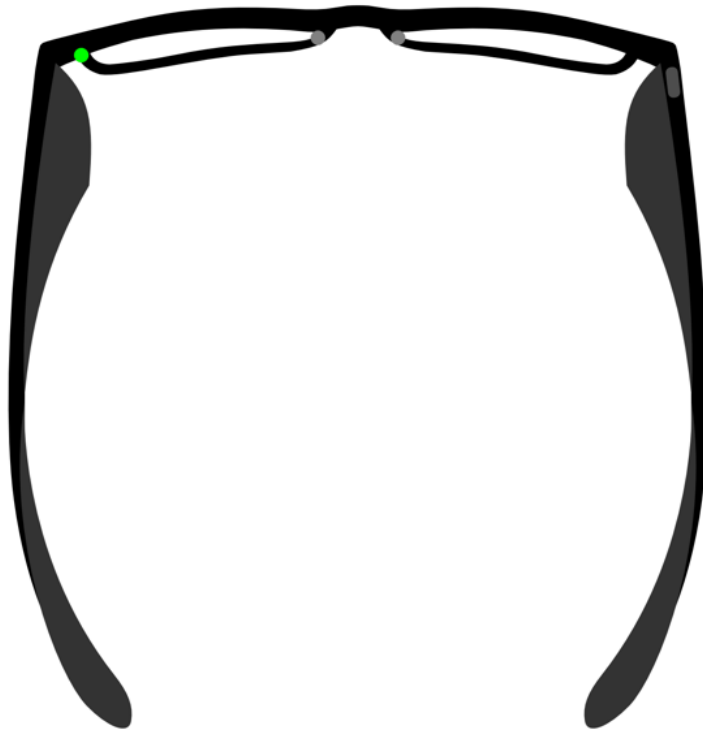


Figure 25 (Poláček, 2020)

Contains:

- LED notification diode (hardware alerts)
- Eye tracking sensors
- Power button
- Temples are thickened in spare place between head and frames, while maintaining foldable construction

***Side View (Temples)***

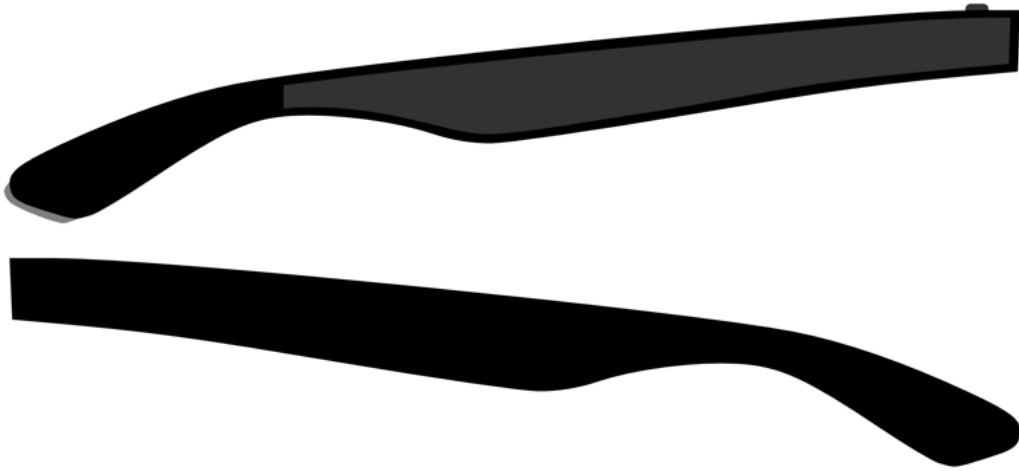


Figure 26 (Poláček, 2020)

Contains:

- Right
  - USB-C socket (hardware boost)
  - Touchpad
  - Power button
- Left
  - –

## 2.3 UI Components

As already mentioned, this UI concept use combination of AR approach and MR possibilities. AR approach is used mostly in 2D system components and MR 3D possibilities in specific apps.

## 2.3.1 General UI Layout

This UI layout focuses mostly on system components. Using AR approach, these elements can be designed on 2D canvas.

### 2.3.1.1 View Zones (2/2)

For the design purposes, I merged areas in *2D Zones (1/2)* to make a proper 2D canvas for smart glasses. Center top and bottom areas have been removed because of impossibility to place content there (e.g. because of merging due to final image composition from both eyes in brain).

#### *2D Zones (2/2)*

Elements placed in dark grey areas are perceived as lateral.



Figure 27 (Poláček, 2020)

By left side



Figure 28 (Poláček, 2020)

By right side



Figure 29 (Poláček, 2020)

#### *3D Zones (2/2)*

Content placed in dark grey areas are in central FoV. Used for 3D elements.



Figure 30 (Poláček, 2020)

### ***Insistent Areas***

Colors and numbers in the zones below describe, how suitable is placing information there.

- 1) This area is the least insistent and at the same time accessible. Always visible content should be placed here.
- 2) This area is the second least insistent. Usually visible content should be placed here.
- 3) This area is the third least insistent. Sometimes visible content should be placed here.
- 4) Content in this area starts to be insistent. Use only for important information.
- 5) Content in this area is insistent. Use for MR mostly.
- 6) Content in this area is very insistent. Use for MR mostly.
- 7) Content placed here is behind inner-frames area. The content there is very inaccessible. Use only for complementary content that the user is not supposed to look at.



Figure 31 (Poláček, 2020)

## 2.3.2 Home

In this UI concept, home is not just the first screen a user sees in the system. It also includes the user's real-world area (room or apartment), where the user lives. Because of this assumption, home section has been divided into 2D home screen using AR approach and 3D home room using MR approach of Immersive apps.

### 2.3.2.1 Home Screen

#### *Low Fidelity Wireframe*

Home screen is basically the transparent area in the lenses. The area is bordered by two compulsory widgets in the corner (Quick Check and Running) and a virtual noticeboard at both sides. The whole area is made of empty slots, where content can be placed.



Figure 32 (Poláček, 2020)

The home screen can be flipped on left side because of left eye dominant users, who represent 30% of population (Halper, 2018).

#### **Noticeboard**

The noticeboard can hold 2D widgets and icons from system or apps. The widgets and apps cannot “float” anywhere at the home screen, they must be attached to the empty slots.



Figure 33 (Poláček, 2020)

### Quick Check

This compulsory widget contains general information about features, that cannot be switched off:

- Time
- Date (month and day)
- Battery status
- Current day in week
- Signal strength
- Phone connection status
- Unread notifications number

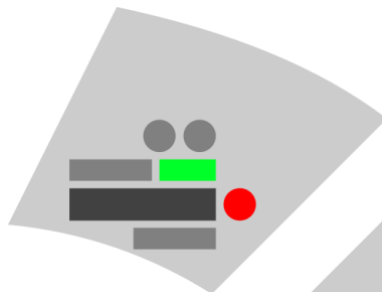


Figure 34 (Poláček, 2020)

### Running

This compulsory widget contains currently running system processes or features (e.g. Wi-Fi, Bluetooth or GPS) and apps (e.g. antivirus or navigation) that can be optionally turned on by the user.

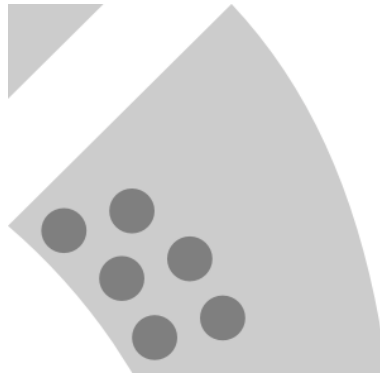


Figure 35 (Poláček, 2020)

### ***High Fidelity Wireframe***

To imitate view of a pair of human eyes, I used widescreen canvas with UXGA standard – 1600 x 1200 pixels dimensions and 4:3 aspect ratio, that should approximate the natural view area (Younis, et al., 2019).

To illustrate empty slots where content will be placed, I set a grid (32 x 32 pixels; 16 divisions; 8 px gutter). The dimensions were set to stay in binary units in order to build create precise wireframe.

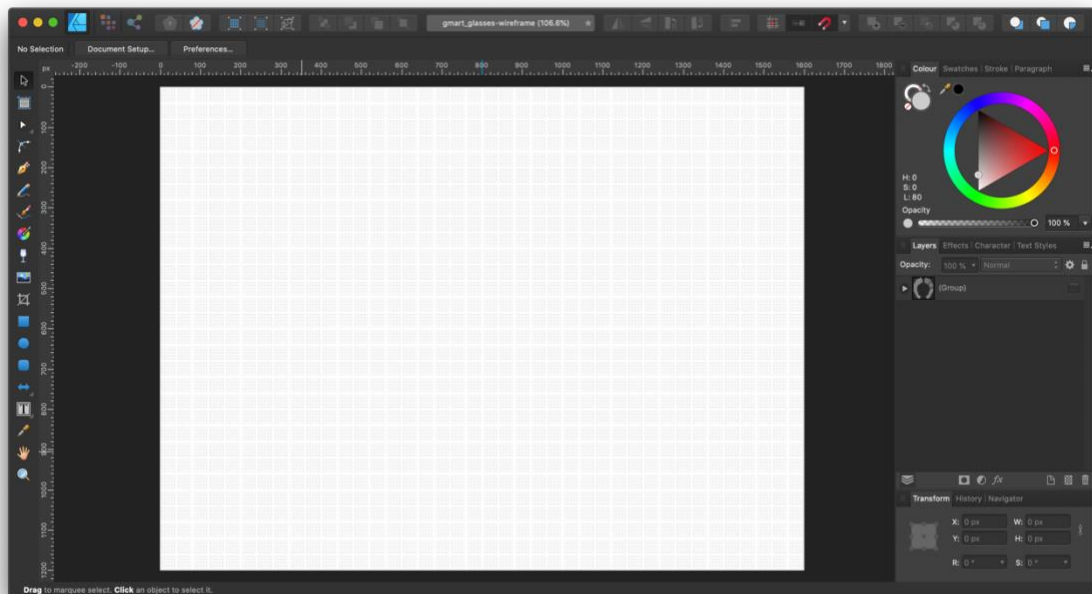


Figure 36 (Poláček, 2020)

Then the grid was merged with low fidelity wireframe scheme and divided to three areas (Quick Check, Running and Noticeboard):

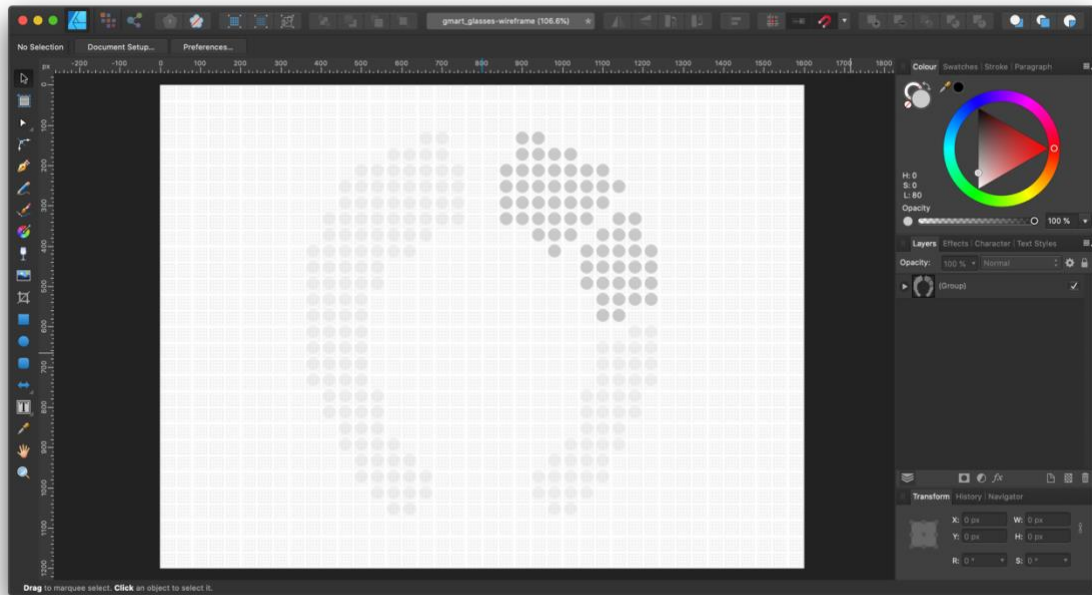


Figure 37 (Poláček, 2020)

In the end the areas were filled with Quick Check, Running and some possible location of widgets and icons inside Noticeboard:



Figure 38 (Poláček, 2020)

### 2.3.2.2 Home Room

At the beginning of the first start of the system, the user is asked to set the user's home room. After setting the room, the user can pin various hologram widgets from their apps

to the user's surroundings, which are always accessible in this location. Similarly, the user can add widgets to other rooms in the house or apartment.

Inspired by Nebula OS of Light by Nreal (Horwitz, 2020)

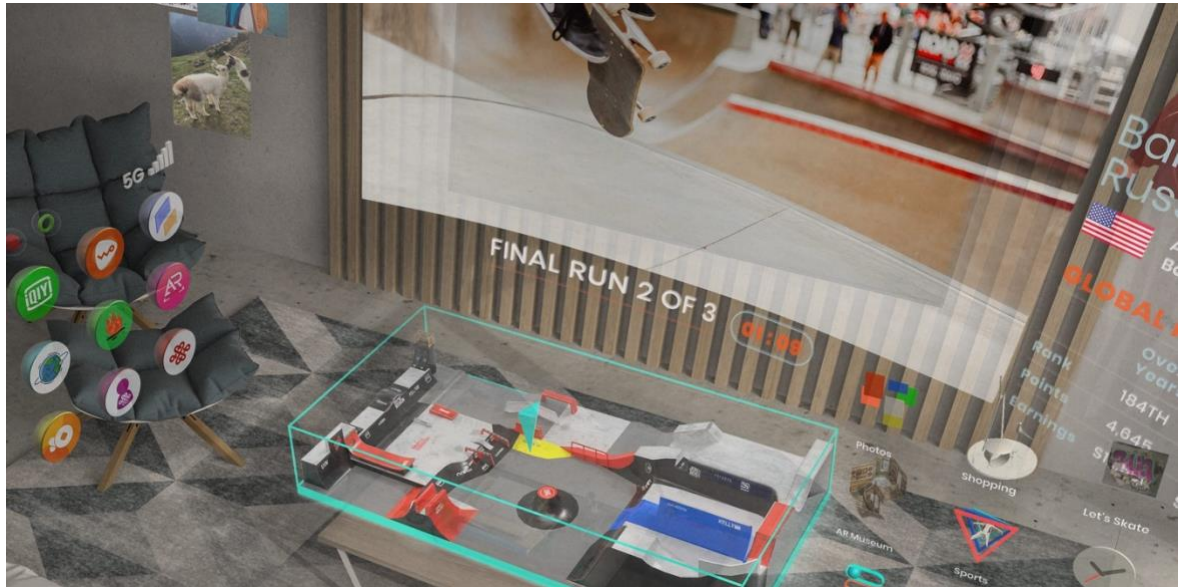


Figure 39 (Horwitz, 2020)

### 2.3.3 Control Center & Settings

Control center is blended together with settings section because control center is simply just a set of pinned settings. Often used settings are included in smart glasses but advanced settings are accessible through a mobile phone.

Control center and settings is accessible on left hand's palm and fingers as 3D immersive content. The settings elements and controls can be selected by touch using fingers of right hand.

Inspired by picture from Gonzales' article (Gonzales).

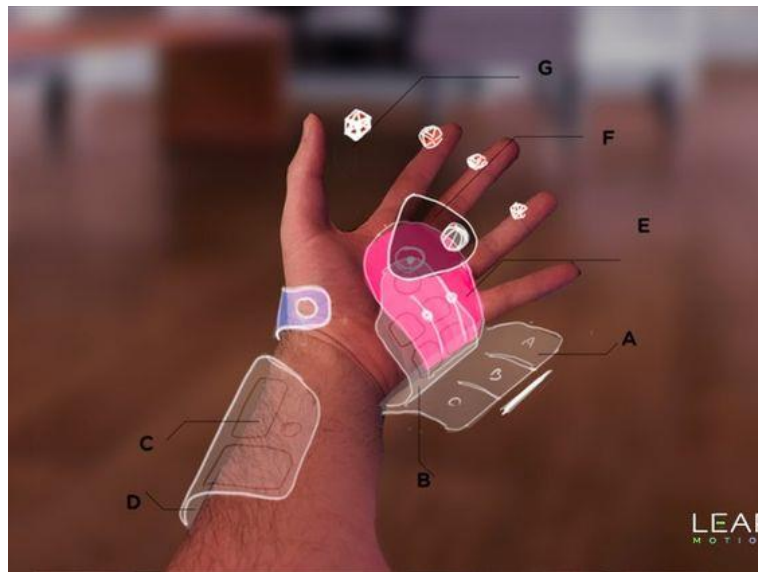


Figure 40 (Gonzales)

Every app also provides its own settings. It is accessible on left hand's wrist right below palm with the same touch input.

Hands themselves provide touch feedback through our skin right into user's brain naturally, so the Feedback Problem is not present here.

## 2.4 Default Apps

This section contains apps, that should be present, when the user starts smart glasses for the first time. There are not third-party apps in the list (e.g. Translator, Facebook or YouTube). Apps are opened inside home screen (something like in Windows OS).

There are apps users know from smart phones, that cannot be operated in smart glasses (e.g. web browser or text-based programs), because their management would be too complicated. However, they can be better managed inside the Gallery

2D (Pictures) & 3D (Holograms) gallery from phone. The objects can be pinned to surrounding in Home Room.

Inspired by Gallery app from Features of One by Magic Leap.

### **Studio**

Simple 2D painting tool and 3D building tool (also for editing objects from Gallery).

Inspired by 3D Viewer from Features of HoloLens by Microsoft and Create app from Features of One by Magic Leap.

Screens app or in a smart phone.

### **Keyboard Feature**

Due to limited ability of text insert, Keyboard appears right where text input is necessary and offers three ways how to do it.

- Written form of what the user says via voice recognition.
- Virtual keyboard hologram in front of the user (on surface or into the air).
- Recognition of a written text it and simultaneously as you write rewrites it to digital sentences.

### 2.4.1 AR & MR Ready Apps

There are apps, that fully use possibilities of AR and MR and operate in user's FoV in the central area:



Figure 41 (Poláček, 2020)

Or, if the app's content should appear only at particular side of the user's view, it can be placed at one these areas (depending on the information's desired side):



Figure 42 (Poláček, 2020)

#### ***Smart Glasses Guide***

A guide that teaches basic gestures and takes the user through the operating system.

Inspired by User Guide app from Features of One by Magic Leap and Learn Gestures app from Features of HoloLens by Microsoft.

#### ***Gallery***

2D (Pictures) & 3D (Holograms) gallery from phone. The objects can be pinned to surrounding in Home Room.

Inspired by Gallery app from Features of One by Magic Leap.

## **Studio**

Simple 2D painting tool and 3D building tool (also for editing objects from Gallery).

Inspired by 3D Viewer from Features of HoloLens by Microsoft and Create app from Features of One by Magic Leap.

## **Screens**

Allows users to project the desktop screen (monitor simulation) to work on with a classic mouse and a keyboard. Connecting a smart phone with a wire may be needed to increase hardware power.

Inspired by Screens app from Features of One by Magic Leap.

## **Meetings**

Feature, that enables user (after turning on) to see another smart glasses' user's avatar in space ahead and collaborate together in supported apps (e.g. create objects in Studio).

Inspired by Social app from Features of One by Magic Leap.

## **Hint**

Shows pinned information that people cannot see you are looking at (e.g. dating hints, speaker notes) and allows to remotely control devices (e.g. presentation). To keep their management invisible they are controlled through natural gestures (e.g. head gestures).

Inspired by Speaker Notes app Features of Focals by North.

## **Driving Companion**

Automatically turns on when the user moves faster than running and disables specific functions that might interrupt attention to driving.

If it is turned on manually, it connects to vehicle's sensors and shows up driving data (e.g. speed or fuel level) in user's insistent view zones to avoid looking out of roadway into periphery zones.

Inspired by Sport Companion app from Features of Glow by MAD Gaze.

## **Camera**

This app captures videos and photos. It is also able to zoom on hardly visible objects (it works like a magnifier or a telescope).

Camera must be controlled in a very visible way to prevent privacy disruption of others. This is a lesson from Glasses by Google failure. There is a LED diode on the glasses that notifies your surrounding that you are capturing your area.

### **Scanner**

This app can scan 3D objects scanning using a surrounding sensor (defining object's shape and dimensions) together with a camera (defining objects texture and color).

Additional feature is scan of QR codes, barcodes or any standardized pattern, which might generate digital content (e.g. website or hologram).

Third feature is OCR text recognition, that basically select written text in front of your eyes and convert it to digital characters (e.g. letters, sentences or whole paragraphs) right into the user's clipboard, available to paste in other apps.

The last feature is classic scanning 2D documents (e.g. images or text) on physical medias (e.g. papers) just as a regular scanner device do. But it can be scanned only by camera, from any angle and right from user's viewpoint.

### **Measure**

User can mark two physical points in surrounding and the app tells, what is the distance between them.

Inspired by AR Ruler from Most Popular AR & MR Apps.

### **Maps**

"Navigation" feature shows the user hologram directions (e.g. where to turn or what to expect) in front of the user's eyes.

Inspired by Google Maps from Most Popular AR & MR Apps.

"Nearby" feature shows places in the user's FoV. The places can show various additional information (e.g. daily offer or opening hours).

Inspired by ViewRanger from Most Popular AR & MR Apps

"Compass" feature shows cardinal points in the user's FoV.

Inspired by Augmented Reality Compass and AR Compass 3D from Most Popular AR & MR Apps.

Other classic features persist in a smart phone or a desktop, because they do not use smart glasses possibilities.

## **Search**

Search app allows to capture an object in front of user's eyes and provide details about the object, found on the internet.

Inspired by Google Lens from Most Popular AR & MR Apps.

It also provides classic search in all zones (e.g. web, documents, phone storage or apps). Other classic features persist in a smart phone or a desktop, because they do not use smart glasses possibilities.

## **Calculator**

Calculates physically written text.

Inspired by MyScript Calculator 2 from Most Popular AR & MR Apps.

Other classic features persist in a smart phone or a desktop, because they do not use smart glasses possibilities.

## **Notes**

“Sticky Notes” feature of Notes app allows the user to type a note and pin it to place, where the note should be visible. Note's visibility to other people can be adjusted for more users (e.g. family or workmates) in sharing preferences.

## **Reminders**

“Sticky Reminders” feature of Reminders app allows the user to type a reminder and pin it to place, where the reminder should trigger the user to carry out the reminder. Reminder's visibility to other people can be adjusted for more users (e.g. family or workmates) in sharing preferences.

## **Weather**

“Night Sky” feature shows images from stars – constellations – on the night sky.

Inspired by Star Walk 2; Big Bang AR; Sky View from Most Popular AR & MR Apps.

“Sun” feature shows sun's following trajectory on the sky and time of sunset and sunrise.

Inspired by Sun Seeker from Most Popular AR & MR Apps

A 2D widget of weather forecast or upcoming rain alert might be useful on a home screen.

Other classic features persist in a smart phone or a desktop, because they do not use smart glasses possibilities.

### ***Health & Sport Companion***

After opening the app it connects to smart watches' sensors and shows up data from there. It might also show up data that helps keeping healthy lifestyle (e.g. steps counter or hydration meter).

Inspired by Sport Companion app from Features of Glow by MAD Gaze.

### ***Translator***

Covers foreign text with translation real-time.

Inspired by Google Translate from Most Popular AR & MR Apps.

## 2.4.2 Non-AR & MR Apps

On the other hand, there are also apps, that do not have to be stored in glasses, because they do not use AR or MR possibilities at all (phone companion apps). Smart glasses can work only as an output device to increase their readiness (it is easier to control them or to see their notifications there), but their full capabilities remains in a smart phone or a desktop.

- **Contacts:** Used to quick contact search.
- **Call:** Used to quick call receiving or dialing.
- **Messages:** Used to show new messages in notifications. Allows quick replies.
- **Mail:** Used to show new mails in notifications. Allows quick replies.
- **Music:** Used for quick songs management (play, pause or next).
- **Recorder:** Used to quickly record a sound or speech.
- **Calendar:** Used to quickly add an event.

## 2.5 Visual Design (2/2)

The visual interface should be based on Skeuomorphic Theory (to make the new platform more accessible) together with Flat Theory (to keep modern look).

### ***Rectangular Sickness***

Computers have been used for a half of century now and screens that people usually perceived were rectangular shape. Smart glasses brought round graphics to the UI and glasses should continue in that philosophy.

## 2.5.1 Feedback Problem

There are currently three possible solutions of the feedback problem:

### 2.5.1.1 Omitting Virtual Buttons

The first is to evade the problem by omitting virtual buttons. All the confirmation would be made through a touchpad on the glasses' temple.

### 2.5.1.2 Using Control Device

The second is to use Control Device with a touchpad to interact with virtual buttons.

### 2.5.1.3 Enhancing Hearing and Sight

The third solution is enhancing hearing and sight feedback in interaction with virtual buttons. Currently in habitual devices (e.g. touch screens, mice or keyboards), the visual feedback is not so important, because there is still a surface that responds to touch. But in MR there is nothing that can respond in the air. So the feedback should be a lot more perceptible by other feasible senses – hearing and sight. (Blokša, 2017)

#### 2.5.1.3.1 Hearing

Desktops do not use sound feedback because mouse and keyboard do it instead. Smart phones use it to simulate mouse and keyboard but it disappears with silent mode. In opposite to smart phones, audio feedback might be ubiquitous (because the temples will sound right to your ears all the time).

The audio should not sound only when button is pressed, but also when is aimed (to simulate “hover” feature that disappeared with touch screen).

#### 2.5.1.3.2 Sight

The visual element of interaction might be enhanced using Skeuomorphic Theory. But while maintaining also Flat Theory, just some skeuomorphic methods should be used.

### ***Plastic Appearance***

The key to a successful visual design of button, that calls to action, is, to define an element, that people naturally tend to touch by sight.

Those are glossy and shaped surfaces. However, two most touchable combinations of surface are glossless, dry and sticky or simple and comfortable. Colors do not affect touch invitation. (Okamoto, et al., 2013)

### ***Natural Animation***

The key to “pressed button” animation, that feels satisfactory, is finding a natural visual response that people see after touching an element. However, users know that the button is not physical so they do not expect touch response from the digital button. So the element should not have touch response.

Despite these conflicting requirements, there are two examples of physical elements without touch response, that also fulfill one of the Plastic Appearance’s condition – glossiness:

- Water – animation of rings around after touch
- Bubble – blows out after touch



Figure 43 (Poláček, 2020)

Created with free stock images by (Chung, 2018) and (Png All, 2018)

## 2.5.2 Colors

The UI should consist of a gradient because of two reasons. Mixing colors inside the gradient might be a metaphor for mixing realities. Moreover, it has a logic purpose – the mix of colors avoids elements to be colorfully blended together with user's surrounding. They are more distinguishable this way. Using artificial colors, that are not usually present in physical world, can increase their recognizability.

The glasses should analyze what colors does the surrounding consist of and then calculate optimal gradient for current view (to stand out of the surrounding). The gradient must be aesthetic at the same time so some kind of color rule should be applied in the calculation.

The text color should be mostly white, because all smart glasses' UIs use it.

### ***Palette***

In search for optimal gradient I got inspired by hologram (or holographic) art, which looks for inspiration in iridescence phenomenon that causes colorful gradient film on some objects' surface (e.g. soap bubble, fuel spill, pearl, ...). (Azzarello, 2016)



Figure 44 (Morgan, 2019)

Holograms are virtual objects, appearing in real environment. This name similarity matches this art style with the MR environment.

I determined three dominant colors using triadic method:

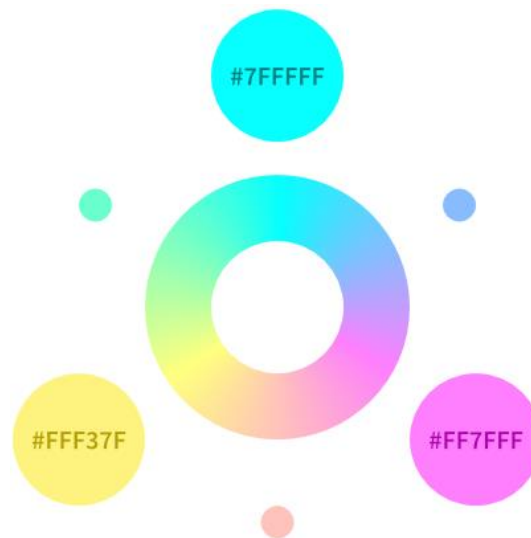


Figure 45 (Poláček, 2020)

- Blue
  - Position: 90°
  - HEX: #7FFFFFFF
- Yellow
  - Position: 165°
  - HEX: #3FFF37F
- Pink
  - Position: 240°
  - HEX: #FF7FFF

Similar color scheme also use Focals by North, which might got to the same conclusion by similar mind flow.

## 2.6 Mockup

The image below shows possible usage of smart glasses UI in real environment. This image applies the gained knowledge from theoretical part and contains the elements I designed in practical part.

There are three areas with different contents:

- Quick Check (top right corner)
- Running (middle right side)
- Noticeboard (right bottom corner; left side)
  - Health & Sport Companion widgets– hydration meter; steps counter
  - Weather widget– incoming weather information
  - Camera widget – telescope to view distant objects

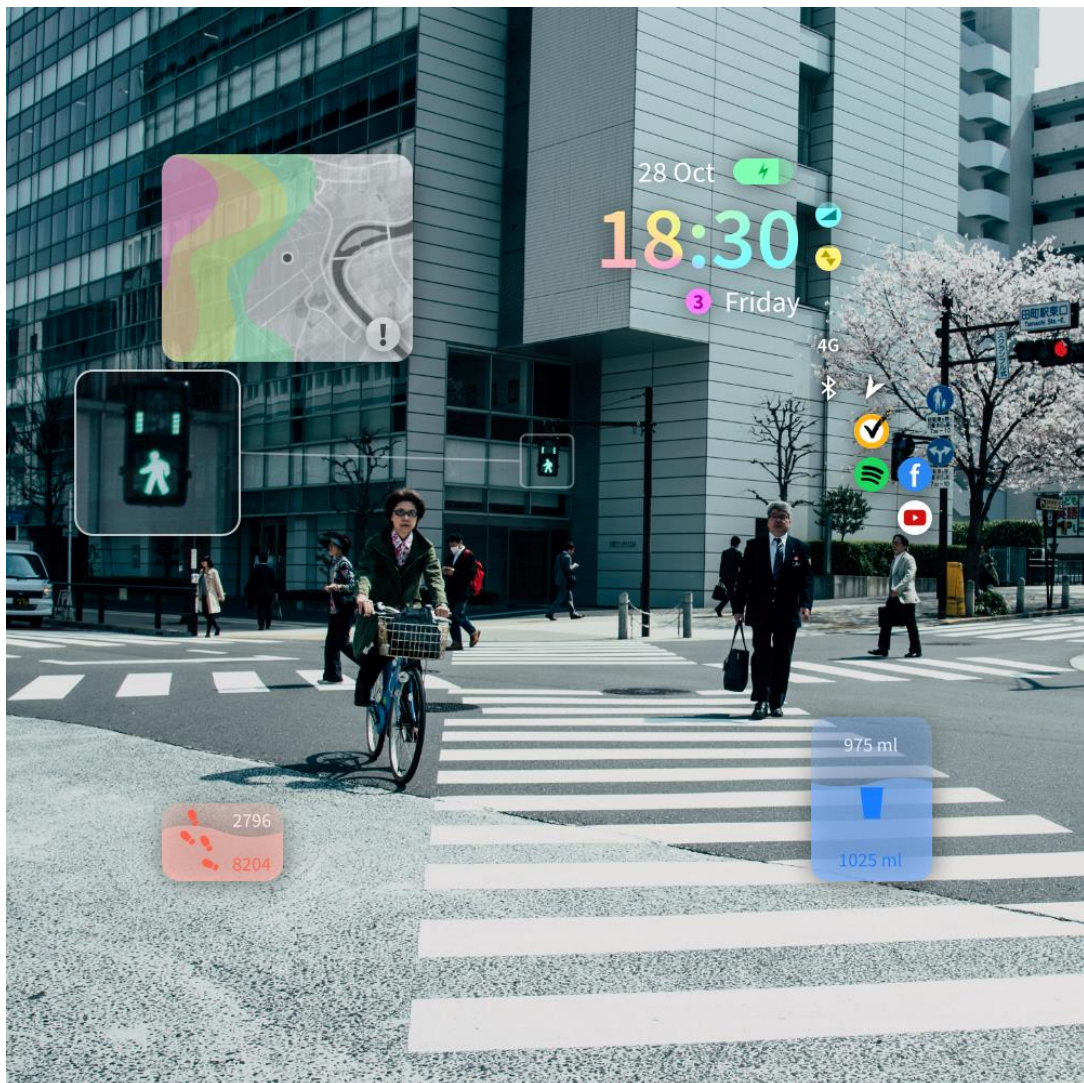


Figure 46 (Poláček, 2020)

Created with free stock images by (Chung, 2018), (Freepik) and (Kissclipart)

## Conclusion

Before I started writing this thesis, I knew just a little about how XR works. While writing this conclusion, I feel like I completely adopted this industry.

I used my analytic skills to map the current industry and to predict future course of smart glasses. I started with articles and manufacturers' developer guides but with need of more experience and visual based information I started to use videos more. With the gained knowledge I designed general UI concept for smart glasses. Unfortunately to this topic, the world of smart glasses turned out to be extremely vast. Although I tried to use Pareto Principle 80/20, this thesis could not cover all the topics of this industry fully. Due to insufficient time that has been allocated to every topic, there is a chance that the conclusions might be incorrect.

This thesis also indicated, what future of digital design will be. The digital world will turn more 3D. There will be no monitors. Smart phones will become smart glasses' companion. Smart watches will be used only for their sensors. And after all, smart glasses might be replaced by smart lenses.

The next step is to upload this thesis to the internet to make it globally accessible. Also, I would like to find some local engineering startup working on a smart glasses prototype, that would allow me to participate in. There I would like to apply my gained knowledge of hybrid UI design.

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